

D. How the Database System Helps Solve the Problem

- 1) The system provides unified data storage through its centralized system which enables complete access to library information without any missing or conflicting records.
- 2) Our system automatically records when a book is borrowed, when it is due back and who borrowed it.
- 3) The system performs automatic fine calculations which guarantee that all calculated fines remain both accurate and equitable.
- 4) The system displays all book statuses by indicating which books are accessible and which ones are currently borrowed or have passed their return date.
- 5) The system enables smoother operations through its book and member search functions which help decrease customer waiting periods.
- 6) The system records every transaction according to its design which establishes an environment of trustworthy records that enable staff members to maintain accountable operations.
- 7) The system produces reports which assist libraries in developing their strategic plans and making operational decisions.

2. Scope and Objectives Scope of the System What the System Will Do

The system will maintain records for books and authors and publishers and book copies.

The system will create library member accounts.

The system will handle the process of borrowing and returning library materials.

The system will automatically track overdue library materials while it calculates fines.

The system will enable users to create reservations.

The system will document all activities performed by staff members.

The system will produce various types of reports.

The system will deliver information to users at high speeds.

What the System Will NOT Do

The system will not provide access to an online public catalogue.

The system will not handle digital content or e-books.

The system will not support payment processing capabilities.

The system will not manage any financial operations or accounting tasks.

The system will not function as a substitute for librarians.

The system will not handle human resources or payroll functions.

System Purpose

The organization will achieve operational improvements through the implementation of automated systems which will eliminate the requirement for manual documentation.

The organization will achieve its objectives through the creation of an accurate system which will store all essential information regarding books and members and their borrowing activities and overdue payments and booking details.

The organization needs to monitor employee behavior while keeping detailed records of all system operations.

The organization will deliver improved services through enhanced search capabilities and faster book borrowing processes which will benefit all users.

The organization will enable better decision-making through report generation as well as statistical analysis which will assist administrators in their resource management.

3. Stakeholders

1. Library Members (Students, Lecturers, Community Users)

Interaction:

- Return books
- Make reservations
- Pay fines for overdue books • View their borrowing history

Benefits:

- Faster access to books
- Accurate records preventing disputes
- Clear information about book availability
- Fair fines calculated automatically

2. Library Staff (Librarians / Assistants)

Interaction:

- Register members
 - Issue and receive books
 - Update reservations and fines
 - Maintain book records
- Benefits:
- Reduced workload
 - Faster data retrieval
 - Fewer errors
 - accountability since actions are recorded
 - Ability to serve many users efficiently

3. Library Management / Administrators

Interaction:

- Review system reports
- Analyse trends for planning
- Supervise staff activity
- Make decisions about inventory and materials

Benefits:

- planning and budgeting
- Clear insights into library trends
- Improved staff monitoring
- Overall enhanced service quality

4. System Administrators / IT Support

Interaction:

- Install. Maintain the system
 - Ensure database security
 - Run. Manage recovery
 - Handle technical issues
- Benefits:
- Easier maintenance due, to clear database structure
 - The database system loss risk leads to data loss. Your data can become corrupted.

- The system will run more reliably because you can depend on it to operate properly.